

Poisson regression – three views

1. Index form

The per-observation reading: what happens for each row i of your data.

$$\begin{aligned} y_i \mid \mu_i &\sim \text{Poisson}(\mu_i) \\ \log(\mu_i) &= \beta_0 + \beta_1 x_i \end{aligned}$$

Coefficient reading. *On the response scale, $\exp(\hat{\beta})$ is the rate ratio: a one-unit increase in the predictor multiplies the expected count by $\exp(\hat{\beta})$ (log link).*

2. Matrix form

The same model in matrix notation. The structural contract every textbook past chapter 4 switches to.

$$\begin{aligned} \mathbf{y} \mid \boldsymbol{\mu} &\sim \text{Poisson}(\boldsymbol{\mu}) \\ \log(\boldsymbol{\mu}) &= \mathbf{X}\boldsymbol{\beta} \end{aligned}$$

3. Worked observation ($i = 1$)

The index-form equations evaluated at the first observation of your data, with coefficient estimates plugged in.

$$\begin{aligned} \hat{\eta}_1 &= 0.955 - 0.0438 \cdot 0.45 \approx 0.936 \\ \hat{\mu}_1 &= \exp(\hat{\eta}_1) = \exp(0.936) \approx 2.55 \\ y_1 &\sim \text{Poisson}(\hat{\mu}_1) \end{aligned}$$

Linear predictor $\hat{\eta}_1 = 0.936$ on the link scale; predicted mean $\hat{\mu}_1 = \exp(\hat{\eta}_1) \approx 2.55$ on the response scale. No additive error term enters the linear predictor – the spread is set by the likelihood above.

Stacking the same response equation for all $n = 100$ observations:

$$\underbrace{\begin{bmatrix} 0.936 \\ 0.956 \\ 0.969 \\ 0.996 \\ 1.02 \\ \vdots \\ 0.963 \\ 0.911 \end{bmatrix}}_{\hat{\boldsymbol{\eta}}_{100 \times 1} \text{ (linear predictor, link scale)}} = \underbrace{\begin{bmatrix} 1 & 0.45 \\ 1 & -0.0186 \\ 1 & -0.318 \\ 1 & -0.929 \\ 1 & -1.49 \\ \vdots & \vdots \\ 1 & -0.166 \\ 1 & 1.02 \end{bmatrix}}_{\mathbf{X}_{100 \times 2}} \underbrace{\begin{bmatrix} 0.955 \\ -0.0438 \end{bmatrix}}_{\hat{\boldsymbol{\beta}}_{2 \times 1} \text{ (estimated)}}$$